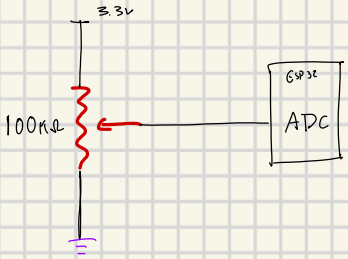
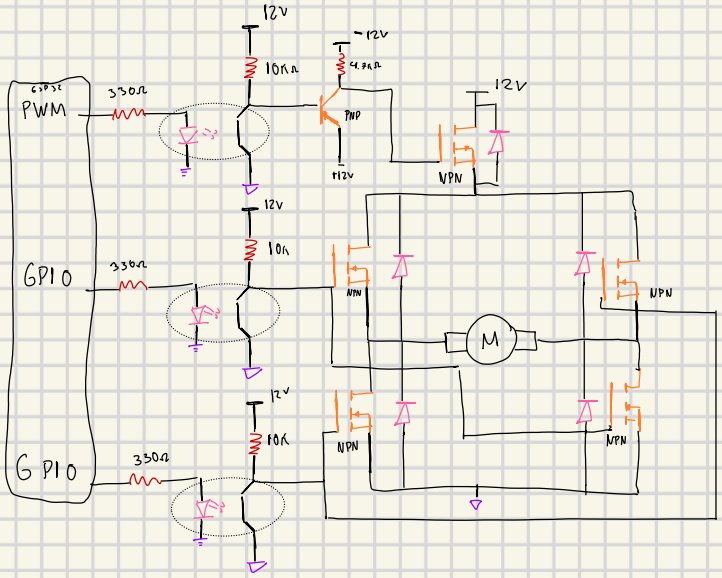
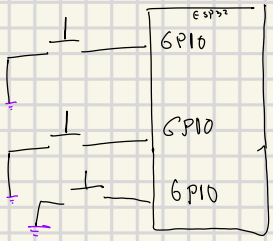
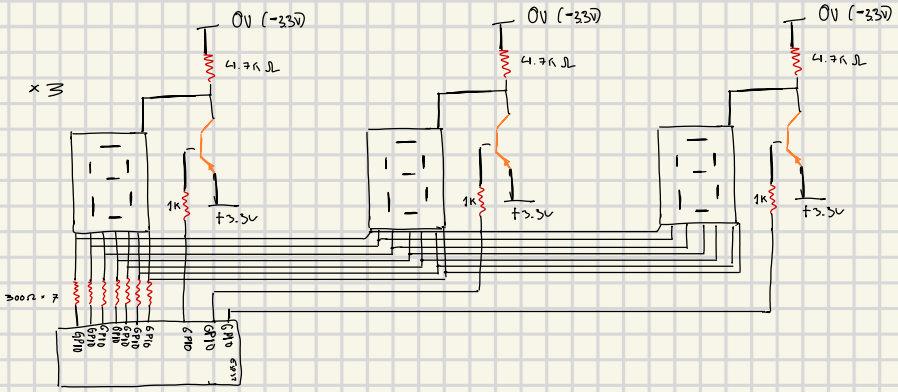

3.3V
100kΩ
ESP32
ADC


ESP32
PWM
GPIO
GPIO
330Ω
330Ω
330Ω
12V
10kΩ
10k
10k
-12V
4.7kΩ
PNP
+12V
12V
NPN
M


ESP32
GPIO
GPIO
GPIO


×3
0V (-3.3V)
4.7kΩ
1k
+3.3V
300Ω × 7
GND
GND